

Bhavdeep Arora

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Education

Toronto Metropolitan University

Bachelor of Science (Honours) Computer Science · Software Engineering Concentration

- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Data Science

Toronto, ON

May 2026

Technical Skills

Languages: Python, Java, C/C++/C#, JavaScript, HTML/CSS, Go, Rust

Frameworks: React, Node.js, Tailwind, Django, Flask, ASP.NET

Developer Tools: Git, Docker, Google Cloud Platform, TravisCI

Data & Cloud: Hadoop, Apache Spark, MySQL, MongoDB, SQLite, AWS, Azure, Firebase

Projects & Research

🌐 bhavdeeparora.dev/projects

Thunderhead | Go, Reverse Proxy, HTTP, Behavioral Analysis | 🌐 Live | 🐙 Github

- Engineered a lightweight reverse proxy in Go that analyzes inbound HTTP traffic across **five behavioral heuristics**—including crawl velocity, robots.txt violations, request frequency, and header anomalies—to identify automated bot activity **without third-party security services**
- Developed a configurable **risk-scoring engine** that assigns request scores from 0–100 across **12** weighted signals and applies graduated mitigations (allow, tarpit, or block), correctly flagging **94%** of automated traffic across **50,000** requests while maintaining a **2%** false-positive rate on legitimate users

Precursor | Python, statsmodels, VAR, Granger Causality, NetworkX | 📄 Paper | 🐙 Github

- Ran an empirical study testing whether commodity futures momentum **Granger-causes** momentum in related sector equity ETFs, using daily data from 2010–2026 across **five commodity-equity pairs**
- Confirmed **all five** hypothesised spillovers at $\alpha = 0.05$ (Gold→GDX: $F=18.90$, $p<0.001$) via Granger causality, VAR, and DAG analysis; used rolling Granger tests, Chow structural break analysis, and a signal-based backtest achieving a **Sharpe ratio of 1.4** to show spillovers are regime-dependent rather than persistent

Groat | Rust, Axum, SQLite, LanceDB, Embeddings | 🌐 Live | 🐙 Github

- Built a **self-hosted, OpenAI-compatible LLM proxy** in Rust (4-crate workspace: server, core logic, provider layer, storage) that cuts API spend **without requiring any application code changes**
- Implemented **semantic response caching** via bge-small-en-v1.5 embeddings and LanceDB similarity search (0.95 threshold), automatic prompt-cache-boundary injection to claim provider-side **50–90%** cached-token discounts, and user-defined cost-tier routing, cutting redundant API spend by **31%** across **8,200** logged requests

Funes | Rust, Ollama, SQLite, Vector Embeddings, CLI | 🌐 Live | 🐙 Github

- Built a **local-first semantic memory system** in Rust that indexes files, notes, and shell history into vector embeddings, enabling natural-language retrieval across a developer's knowledge base **without cloud infrastructure**
- Architected a background indexing daemon with filesystem monitoring, chunking pipelines, and SQLite-backed vector storage; integrated local embedding and generation models to return relevant results in **85ms** on average across **3,400** indexed documents, achieving **91%** top-5 retrieval accuracy

Lacunae | Python, PyTorch, U-Net, fastMRI, HDF5 | 🐙 Github

- Trained a **7.7M-parameter U-Net** on ~100 GB of fastMRI single-coil knee MRI data to reconstruct **diagnostic-quality images** from $8\times$ undersampled k-space, achieving **SSIM: 0.7803** and **PSNR: 30.8 dB** on held-out validation scans—an improvement of **18%** SSIM over zero-filled baseline reconstruction

Verrere | Next.js 15, TypeScript, Prisma, PostgreSQL, Clerk, Hardcover GraphQL | 🌐 Live | 🐙 Github

- Built a full-stack book discovery platform delivering **personalized recommendations** via a swipe-based interface, integrating the Hardcover GraphQL API with PostgreSQL/Prisma persistence and achieving a **38%** swipe-to-save rate across **120** active users

Leadership & Volunteer Experience

Data for Good Toronto | Data Analyst

May 2026 – Present

- Collaborate with a cross-functional team of analysts, engineers, and stakeholders to evaluate datasets for non-profit and social impact organizations, supporting evidence-based decision making
- Develop data-cleaning, transformation, and exploratory analysis workflows in Python and pandas, contributing to end-to-end analytics pipelines for community-focused projects

SciXchange | Technical Assistant

September 2023 – April 2024

- Guided 50+ visitors through interactive science exhibits, translating technical STEM concepts into engaging and accessible explanations for diverse audiences
- Supported hands-on educational activities and event operations, fostering participant engagement and scientific curiosity through public outreach initiatives

Hobbies: Reading Fiction, Trekking, Weightlifting, Woodworking · **Interests:** Mathematics, Environmental Engineering, Quantum Mechanics, Astrophysics